DSC 243 — Homework 2

The Positive Semidefinite Case and Convergence Under Spectral Structure

Instructions. Use the notation from the notes. Unless explicitly stated otherwise, assume

$$\begin{aligned}
&f(x) = \tfrac{1}{2}x^\top Ax - b^\top x,\\
&A = A^\top \succeq 0,\\
&b \in \operatorname{range}(A),\\
&\beta = \lambda_{\max}(A),\\
&\alpha = \lambda_{\min}(A),\\
&\text{and when } \alpha > 0,\ \kappa = \beta/\alpha.
\end{aligned}$$

Let $x^* \in \{x : Ax = b\}$ and $e_0 = x_0 - x^*$. Let $v_1, \dots, v_d$ be an orthonormal eigenbasis of A with eigenvalues $\lambda_1, \dots, \lambda_d$, and write $e_0 = \sum_{i=1}^d c_i v_i$. Throughout, we repeatedly use the exact spectral formula

$$f(x_k) - f^* \;=\; \tfrac{1}{2}\sum_{i=1}^{d} \lambda_i\, p_k(\lambda_i)^2\, c_i^2, \qquad p_k(\lambda) := \prod_{j=1}^{k}(1-\eta_j\lambda),$$

derived in Section 2 for gradient descent with time-varying stepsizes $\eta_1, \dots, \eta_k$. You may quote results proved in class or in the notes, but every nontrivial step must be justified.

Part I: Theory

T1. Sublinear convergence of GD in the positive semidefinite case. In this problem you will carefully derive the $O(1/k)$ rate of Theorem 6.1 and extract some consequences.

(a) Prove the elementary inequality

$$\sup_{t\in[0,1]} t(1-t)^q \;=\; \frac{1}{q+1}\left(\frac{q}{q+1}\right)^q \;\le\; \frac{1}{q+1} \qquad \text{for every } q > 0,$$

and identify the point $t^* \in [0,1]$ at which the supremum is attained.

(b) Using (a), prove *Theorem 6.1*: with constant stepsize $\eta = 1/\beta$,

$$f(x_k) - f^* \;\le\; \frac{\beta}{2(2k+1)}\,\|x_0 - x^*\|^2.$$

You may use the spectral formula stated in the instructions above.

(c) *Sharper constant.* Show that the argument in (b) in fact gives

$$f(x_k) - f^* \;\le\; \frac{\beta}{2(2k+1)}\left(\frac{2k}{2k+1}\right)^{2k}\|x_0 - x^*\|^2.$$

What is the limit of $(2k/(2k+1))^{2k}$ as $k \to \infty$? Interpret the improvement over (b) for large k.

(d) *General constant stepsize.* Fix any $\eta \in (0, 1/\beta]$ and let k be large enough that $1/(2k+1) \le \eta\beta$. By repeating the argument of (b) using the substitution $t = \eta\lambda$, show that

$$f(x_k) - f^* \;\le\; \frac{1}{2\eta(2k+1)} \|x_0 - x^*\|^2.$$

Among $\eta \in (0, 1/\beta]$, which value minimizes this upper bound, and what does it reduce to?

(e) *Iteration complexity.* Using (b), determine the smallest k that guarantees

$$f(x_k) - f^* \;\le\; \varepsilon,$$

as a function of β, $\|x_0 - x^*\|^2$ and ε.

T2. Chebyshev polynomials of the second kind and PSD acceleration. Define U_j by $U_0(x) = 1$, $U_1(x) = 2x$, and the recurrence

$$U_{j+1}(x) = 2xU_j(x) - U_{j-1}(x) \qquad (j \ge 1).$$

(a) *Trigonometric form.* Show by induction that

$$U_j(\cos\theta) \;=\; \frac{\sin\big((j+1)\theta\big)}{\sin\theta}, \qquad \theta \in (0, \pi).$$

(b) Compute $U_j(1)$ and $U_j(-1)$. Show that U_j has j simple roots in $(-1, 1)$ at $\cos(\ell\pi/(j+1))$ for $\ell = 1, \ldots, j$. Deduce that U_j has leading coefficient 2^j.

(c) *Orthogonality.* Show that with respect to the weight $\sqrt{1-x^2}$ on $[-1, 1]$,

$$\int_{-1}^{1} U_j(x)\, U_k(x)\, \sqrt{1-x^2}\, dx \;=\; \frac{\pi}{2}\, \delta_{jk}.$$

Hint: substitute $x = \cos\theta$ and use the trigonometric form.

(d) *Weighted extremal property.* For any polynomial p of degree exactly k with the same leading coefficient as U_k (that is, 2^k), show that

$$\max_{x\in[-1,1]} \sqrt{1-x^2}\, |p(x)| \;\ge\; \max_{x\in[-1,1]} \sqrt{1-x^2}\, |U_k(x)| \;=\; 1.$$

Hint: The identity $\sqrt{1-x^2}\, U_k(x) = \sin\big((k+1)\theta\big)$ for $x = \cos\theta$ shows that $\sqrt{1-x^2}\, U_k(x)$ attains ± 1 at $k+1$ alternating values of x. Consider the polynomial $r := p - U_k$ of degree at most $k-1$, and count the sign changes of $\sqrt{1-x^2}\, r$ forced on you by the assumption that $\sqrt{1-x^2}|p| < 1$ everywhere.

(e) *Proof of Lemma 6.2.* Let $\phi_k(\lambda) := \big(1 - \frac{\lambda}{\beta}\big) U_{k-1}(1 - 2\lambda/\beta)/k$. Using the substitution $\cos\theta = 1 - 2\lambda/\beta$, show that

$$\max_{\lambda\in[0,\beta]} \lambda\, \phi_k(\lambda)^2 \;\le\; \frac{\beta}{4k^2}.$$

(f) *Proof of Theorem 6.2.* Verify that the stepsizes $\eta_j = \big(\beta \sin^2(j\pi/(2k))\big)^{-1}$ for $j = 1, \ldots, k$ produce the polynomial $p_k = \phi_k$ in the spectral formula, and deduce the PSD Chebyshev rate

$$f(x_k) - f^* \;\le\; \frac{\beta}{8k^2}\, \|x_0 - x^*\|^2.$$

Compare the iteration complexity with that of T1(e).

T3. Source condition and the $O(k^{-(1+2s)})$ rate of GD. Suppose the initial error satisfies the source condition

$$e_0 \;=\; A^s\, w \qquad \text{for some } w \in \mathbb{R}^d \text{ with } \|w\|^2 \le M \text{ and } s > -\tfrac{1}{2}.$$

(a) Writing $c_i = \lambda_i^s \tilde{c}_i$ with $\tilde{c}_i := v_i^\top w$, substitute into the spectral formula (with $\eta = 1/\beta$) to obtain

$$f(x_k) - f^* \;\le\; \frac{\|w\|^2}{2} \max_{\lambda \in [0,\beta]} \lambda^{1+2s}(1-\lambda/\beta)^{2k}.$$

(b) Setting $t = \lambda/\beta$, show that

$$\max_{t\in[0,1]} t^{1+2s}(1-t)^{2k} \;=\; \left(\frac{1+2s}{2k+1+2s}\right)^{1+2s} \left(\frac{2k}{2k+1+2s}\right)^{2k}.$$

Deduce

$$f(x_k) - f^* \;\le\; \frac{\beta^{1+2s}}{2} \left(\frac{1+2s}{2k+1+2s}\right)^{1+2s} \|w\|^2,$$

which is Theorem 7.1.

(c) Verify that at $s = 0$ the bound of (b) recovers (the slightly sharper form of) Theorem 6.1 derived in T1(c).

(d) Show that the asymptotic rate is

$$f(x_k) - f^* \;=\; O\big(\beta^{1+2s}\, k^{-(1+2s)}\, \|w\|^2\big) \qquad \text{as } k \to \infty.$$

(e) *Iteration complexity.* For $s > -\frac{1}{2}$, derive the smallest k (up to constants) such that $f(x_k) - f^* \le \varepsilon$. In particular, how does the complexity blow up as $s \downarrow -\frac{1}{2}$?

(f) *Interpretation.* Explain in one or two sentences why the source condition improves the exponent from -1 (Theorem 6.1) to $-(1+2s)$, in terms of which eigendirections carry most of the initial error.

T4. Power-law spectral density: the GD rate. Assume the spectral error measure $\mu = \sum_i c_i^2 \delta_{\lambda_i}$ is well-approximated by a continuous density $d\mu(\lambda) = \phi(\lambda)\, d\lambda$ with

$$\phi(\lambda) \;=\; M\, \lambda^{a-1} \qquad \text{on } (0,\beta], \qquad M > 0,\; a > -1,$$

and define the idealized spectral-integral error

$$\mathcal{E}_k \;:=\; \frac{1}{2} \int_0^\beta \lambda\, p_k(\lambda)^2\, \phi(\lambda)\, d\lambda.$$

Take $p_k(\lambda) = (1-\lambda/\beta)^k$, corresponding to GD with $\eta = 1/\beta$.

(a) Show that with the substitution $t = \lambda/\beta$,

$$\mathcal{E}_k \;=\; \frac{M\,\beta^{a+1}}{2} \int_0^1 t^a (1-t)^{2k}\, dt \;=\; \frac{M\,\beta^{a+1}}{2}\, B(a+1,\, 2k+1),$$

where $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ is the Beta function. Deduce

$$\mathcal{E}_k \;=\; \frac{M\,\beta^{a+1}}{2} \cdot \frac{\Gamma(a+1)\,\Gamma(2k+1)}{\Gamma(2k+a+2)}.$$

(b) Using the standard estimate $\Gamma(n+c)/\Gamma(n) \sim n^c$ as $n \to \infty$ (which follows from Stirling), prove the asymptotic

$$\mathcal{E}_k \;\sim\; \frac{M\,\Gamma(a+1)\,\beta^{a+1}}{2\,(2k)^{a+1}} \qquad \text{as } k \to \infty.$$

(c) Verify the two special cases: $a = 1$ (flat spectrum $\phi = M$ on $(0, \beta]$, constant eigenvalue density) gives $O(k^{-2})$; the critical exponent $a = 1/2$ gives $O(k^{-3/2})$.

(d) *Comparison with the vanilla $O(1/k)$ bound.* Compare the rate $O(k^{-(a+1)})$ of (b) with the worst-case Theorem 6.1 bound $O\big(\beta\,\|e_0\|^2/k\big)$. In which range of a is (b) strictly better (same or stronger rate *and* finite constant, even when $\|e_0\|^2$ is infinite)? Justify.

(e) *Source condition as a spectral integral.* Suppose the eigenvalue density of A behaves polynomially, $\rho_A(\lambda) \propto \lambda^{-1/\mathsf{a}-1}$ near $\lambda = 0$, and the coefficients $\tilde{c}_i = v_i^\top w$ from T3 are isotropic. Show that the effective spectral error density is $\phi(\lambda) \propto \lambda^{2s-1/\mathsf{a}-1}$, so that the power-law exponent is $a = 2s - 1/\mathsf{a}$. Use (b) to deduce the resulting GD rate in terms of s and a. For what ranges of s and a is the resulting rate faster than $O(1/k)$?

T5. (Optional) CG on Marchenko–Pastur at $\gamma = 1$. At the critical aspect ratio $\gamma = 1$ the Marchenko–Pastur law has no atom at zero and its density is

$$\rho_{MP}(\lambda) \;=\; \frac{\sqrt{4-\lambda}}{2\pi\sqrt{\lambda}} \qquad \text{on } [0,4].$$

Throughout this problem you may take for granted Theorem 7.4 from the notes ("minimum-norm polynomial via a reproducing kernel") as well as the following explicit form of its conclusion: if $\{\psi_j\}_{j=0}^k$ is an orthonormal basis of $\mathcal{P}_k$ with respect to the inner product $\langle f, g\rangle = \int fg\,d\mu$ and $x_0 \in \mathbb{R}$ is fixed, then

$$\min_{\substack{p\in\mathcal{P}_k\\ p(x_0)=1}} \int p(\lambda)^2\,d\mu(\lambda) \;=\; \frac{1}{\sum_{j=0}^k \psi_j(x_0)^2}. \tag{21}$$

We take the spectral assumption of the notes: the reweighted empirical measure satisfies $\nu_d \Rightarrow \|e_0\|^2\mu_{MP}$, which reduces the analysis of CG at $\gamma = 1$ to the polynomial problem

$$\mathcal{E}_k \;=\; \frac{\|e_0\|^2}{2} \min_{\substack{p\in\mathcal{P}_k\\ p(0)=1}} \int_0^4 p(\lambda)^2\,d\mu(\lambda), \qquad d\mu(\lambda) := \lambda\,\rho_{MP}(\lambda)\,d\lambda.$$

(a) Show that $d\mu(\lambda) = \dfrac{\sqrt{\lambda(4-\lambda)}}{2\pi}\,d\lambda$ on $[0,4]$.

(b) Apply the affine change of variables $x(\lambda) := \lambda/2 - 1$, mapping $[0,4] \to [-1,1]$. Show that in the new coordinate,

$$d\mu \;=\; \frac{2}{\pi}\sqrt{1-x^2}\,dx.$$

(c) Using T2(c) and the formula $d\mu = (2/\pi)\sqrt{1-x^2}\,dx$ from (b), verify that

$$\psi_j(\lambda) \;:=\; U_j\big(x(\lambda)\big), \qquad j = 0, 1, \ldots, k,$$

is an *orthonormal* basis of $\mathcal{P}_k$ in $L^2(\mu)$. In particular, show $\int \psi_j\psi_\ell\,d\mu = \delta_{j\ell}$.

(d) Using $U_j(-1) = (-1)^j(j+1)$, show that $\psi_j(0)^2 = (j+1)^2$.

(e) Using (a)–(d) and formula (21) at $x_0 = 0$, show

$$\min_{\substack{p\in\mathcal{P}_k\\ p(0)=1}} \int_0^4 p(\lambda)^2\, d\mu(\lambda) \;=\; \frac{1}{\sum_{j=0}^{k}(j+1)^2} \;=\; \frac{6}{(k+1)(k+2)(2k+3)}.$$

(f) Deduce the *CG on Marchenko–Pastur* bound

$$\mathcal{E}_k \;=\; \frac{3\,\|e_0\|^2}{(k+1)(k+2)(2k+3)} \;=\; O\big(\|e_0\|^2/k^3\big).$$

(g) Compare this with the GD rate $\mathcal{E}_k \sim \|e_0\|^2/(\sqrt{2\pi}\,k^{3/2})$ at $\gamma = 1$ (derived by specializing T4(b) to $a = 1/2$ and $M = \|e_0\|^2/\pi$). By what factor does CG outperform GD for $\varepsilon = 10^{-6}$?

Part II: Experiments

E1. GD, PSD Chebyshev, and CG on a power-law spectrum. Take $d = 200$ and

$$A \;=\; \mathrm{diag}\big(i^{-3}\big)_{i=1}^{d}$$

(so $A \succ 0$ with $\alpha = d^{-3}$, $\beta = 1$ and $\kappa = d^3 \approx 8\times 10^6$). Let $x^* = \mathbf{1} \in \mathbb{R}^d$, $b = Ax^*$. Use the same generic random initial point x_0 for every method.

(a) Run gradient descent with constant stepsize $\eta = 1/\beta$ for $k = 5000$ iterations.

(b) For each horizon $K \in \{10, 20, 50, 100, 200, 500, 1000, 2000\}$, run PSD-Chebyshev with the stepsize schedule $\eta_j = 1/(\beta \sin^2(j\pi/(2K)))$ for $j = 1, \ldots, K$, starting from the same x_0. Record the final $f(x_K) - f^*$ for each choice of K and plot these values as a function of K.

(c) Run the conjugate gradient method from x_0 for $k = 5000$ iterations (truncating once the gap drops below $10^{-13}\|x_0 - x^*\|^2$).

(d) On the same semilog plot of $(f(x_k) - f^*)/(f(x_0) - f^*)$ versus k, overlay:

- the GD curve from (a);
- the final-iterate values from (b), plotted as a step function or marker plot;
- the CG curve from (c);
- the theoretical upper bounds $\beta/(2(2k+1)) \cdot \|x_0 - x^*\|^2/(f(x_0) - f^*)$ (Theorem 6.1) and $\beta/(8k^2) \cdot \|x_0 - x^*\|^2/(f(x_0) - f^*)$ (Theorem 6.2).

(e) Comment on the empirically observed rates of GD, PSD-Chebyshev, and CG. Which is fastest? For large k, does CG separate from PSD-Chebyshev, and if so, why? (Recall that the spectrum here decays polynomially, so the spectral-density analysis of Section 7 is relevant.)

E2. Source condition and the matching GD rate. Take $d = 2000$ and

$$A \;=\; \mathrm{diag}\big(i^{-\mathsf{a}}\big)_{i=1}^{d} \qquad \text{with } \mathsf{a} = 2.$$

For each $s \in \{-\frac{1}{4}, 0, \frac{1}{2}, 1\}$, generate an initial error satisfying the source condition $e_0 = A^s w$ as follows: sample $w \in \mathbb{R}^d$ with iid $\mathcal{N}(0,1)$ entries (rescale so $\|w\|^2 = d$), and set $x_0 = x^* + e_0$ with $x^* = \mathbf{1}$, $b = Ax^*$.

(a) For each s, run GD with $\eta = 1/\beta$ for $k = 10^4$ iterations. Plot $(f(x_k) - f^*)/(f(x_0) - f^*)$ versus k on a log-log scale (four curves, one per value of s).

(b) On each curve, overlay the theoretical reference line $c_s\, k^{-(1+2s)}$ (dotted), where c_s is the theoretical constant from Theorem 7.1 (expressed in terms of β, s, $\|w\|^2$, $f(x_0) - f^*$). Do the empirical slopes match the theoretical exponent $-(1 + 2s)$?

(c) For $s = -1/4$ (the case with $s < 0$), compare the empirical value of $\|e_0\|^2$ and $f(x_0) - f^*$. Observe that $\|e_0\|^2$ is large but the function gap remains well-defined. Explain in one or two sentences how Theorem 7.1 delivers a *dimension-free* bound in this regime (that is, the right-hand side of the bound depends on $\|w\|$, not $\|e_0\|$).

(d) Compute iteration complexity empirically: for each s, report the smallest k that achieves relative gap 10^{-4} and 10^{-6}. Does the ratio of complexities across s match what T3(e) predicts?

E3. CG on Marchenko–Pastur spectra across the three regimes. Fix $d = 1200$ and consider three aspect ratios $\gamma \in \{0.5, 1, 2\}$:

$$n_{0.5} \;=\; 2d, \qquad n_1 \;=\; d, \qquad n_2 \;=\; d/2.$$

For each γ, draw $D \in \mathbb{R}^{n\times d}$ with iid $\mathcal{N}(0,1)$ entries and set $A = \frac{1}{n}D^\top D$. Sample x^* uniformly on the sphere of radius $\sqrt{d}$ and set $b = Ax^*$, $x_0 = 0$. Run both GD with stepsize $\eta = 1/\lambda_{\max}(A)$ and CG, and record $f(x_k) - f^*$ versus k. For each γ, take the *median* over 20 independent draws of D (and x^*).

(a) For each γ, plot on a log-log scale the median relative gap $(f(x_k) - f^*)/(f(x_0) - f^*)$ versus k for GD and CG. Shade the 10%–90% interquantile range.

(b) On the $\gamma = 1$ plot, overlay the dotted reference lines

$$\text{GD reference:} \quad \frac{1}{\sqrt{2\pi}\, k^{3/2}} \cdot \frac{\|e_0\|^2}{f(x_0) - f^*},$$

$$\text{CG reference (from T5(f)):} \quad \frac{3}{(k+1)(k+2)(2k+3)} \cdot \frac{\|e_0\|^2}{f(x_0) - f^*}.$$

Report whether the empirical rates track the predicted $k^{-3/2}$ and k^{-3} exponents.

(c) On the $\gamma = 0.5$ plot (positive definite case), overlay the *theoretical Chebyshev upper bound*

$$4\left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^{2k} \cdot 1, \qquad \kappa \;=\; \big((1 + \sqrt{\gamma})/(1 - \sqrt{\gamma})\big)^2,$$

and also a power-times-exponential reference $c\, k^{-3/2}(1 - 1/\kappa)^{2k}$ (from the Laplace-method Theorem 7.3 applied to the MP edge). Which of the two is closer to empirical CG for early k? Late k? Briefly explain why.

(d) On the $\gamma = 2$ plot (rank-deficient), verify that in the effective spectrum the *nonzero* eigenvalues lie in $[(\sqrt{\gamma}-1)^2, (\sqrt{\gamma}+1)^2]$. Overlay the PSD Chebyshev/CG worst-case bound $\beta/(8k^2) \cdot \|e_0\|^2$ as a reference, as well as the power-times-exponential reference from the Laplace method with $\alpha_{\text{eff}} = (\sqrt{\gamma} - 1)^2$ and $\beta = (\sqrt{\gamma} + 1)^2$. Discuss whether the effective-condition-number rate or the power-times-exponential rate better describes empirical behavior.

(e) *Summary.* In one or two sentences, describe how the three MP regimes ($\gamma < 1$, $\gamma = 1$, $\gamma > 1$) differ from the point of view of convergence of CG: where does the $k^{-3/2}$ edge factor come from, and in which regime does CG achieve the sharpest gain over GD?